

Feature Selection Report

Customer Personality Segmentation Project

Module 05 — Feature Engineering & Data Preprocessing (CPS-M05)

1. Objective

This report documents which features were removed or retained ahead of customer segmentation (clustering), and the reasoning behind each decision. The goal is a feature set that is behaviourally meaningful, non-redundant, and fully numeric.

2. Selection Summary

Feature(s)	Decision	Justification
ID	Removed	Unique identifier; carries no behavioural signal for clustering.
Year_Birth	Removed	Fully replaced by the derived Customer_Age feature.
Age (cleaning-stage)	Removed	Duplicate of the newly created Customer_Age.
Dt_Customer	Removed	Raw date replaced by Customer_Tenure (days since enrollment).
Enrollment_Year/Month/Day	Removed	Superseded by the more directly useful Customer_Tenure.
Kidhome / Teenhome	Removed	Combined into Total_Children to reduce redundancy.
Education (text)	Removed	Replaced by Education_Encoded (ordinal label encoding).
Marital_Status (text)	Removed	Replaced by one-hot Marital_* columns.
Preferred_Shopping_Channel (text)	Removed	Replaced by one-hot Channel_* columns.
Product_Preference (text)	Removed	Replaced by one-hot Product_* columns.
Customer_Activity_Level (text)	Removed	Replaced by Customer_Activity_Level_Encoded.
Z_CostContact / Z_Revenue	Removed	Constant columns (zero variance); no discriminative value.
Income, Recency, Total_Spending, Total_Purchases, Mnt*, Num*Purchases, Customer_Age, Customer_Tenure, Family_Size, Total_Children, Digital_Engagement, Deal_Dependency, Total_Campaign_Acceptance, Avg_Spending_Per_Purchase, encoded categoricals	Kept	Core demographic and behavioural drivers directly relevant to customer segmentation.

3. Outcome

Starting from 33 cleaned columns plus 13 engineered and 5 encoded-categorical groups, 16 columns were removed as identifiers, redundant raw versions, or zero-variance constants. The resulting selected feature set (48 columns prior to transformation) retains only features with direct relevance to purchasing behaviour, engagement, demographics, and campaign responsiveness.

4. Why These Features Matter for Segmentation

- Spending and purchase-count features (Total_Spending, Total_Purchases, Mnt* columns) directly capture customer value — the primary axis for value-based segments.

- Engagement features (Digital_Engagement, Deal_Dependency, NumWebVisitsMonth) capture how and why customers shop, supporting behavioural segments.
- Demographic features (Income, Customer_Age, Family_Size, Total_Children, Education_Encoded) support life-stage-based segments.
- Campaign and Recency features (Total_Campaign_Acceptance, Recency, Customer_Activity_Level_Encoded) support engagement/churn-risk segments.